Lab Report: Running and Evaluating LLMs

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1 Overview

This report details the evaluation of Large Language Models (LLMs) on the MMLU benchmark using both local resources and Google Colab. It compares accuracy and timing across model sizes and details the implementation of a memory-managed Chat Agent.

2 Part 1: Local Evaluation (Small Models)

The evaluation script was run locally on three small-scale models ($\leq$ 1B parameters).

2.1 Results

The Llama 3.2 1B model served as the baseline, achieving the highest accuracy among the small models at 41.8%. The smallest model, SmolLM2-360M, struggled significantly with the complex MMLU tasks.

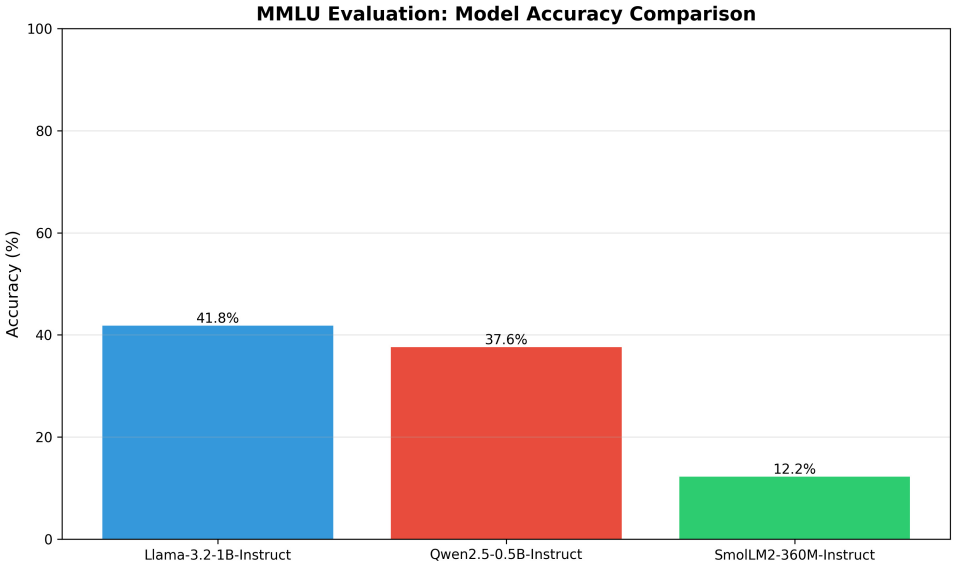


Figure 1: Accuracy comparison of small models running locally.

2.2 Error Analysis

An analysis of question difficulty reveals that errors were not random. As shown in Figure 2, a significant portion of questions were missed by *all three* models. This suggests that certain MMLU subjects (like Professional Law and Math) are fundamentally too difficult for models in this weight class, regardless of architecture.

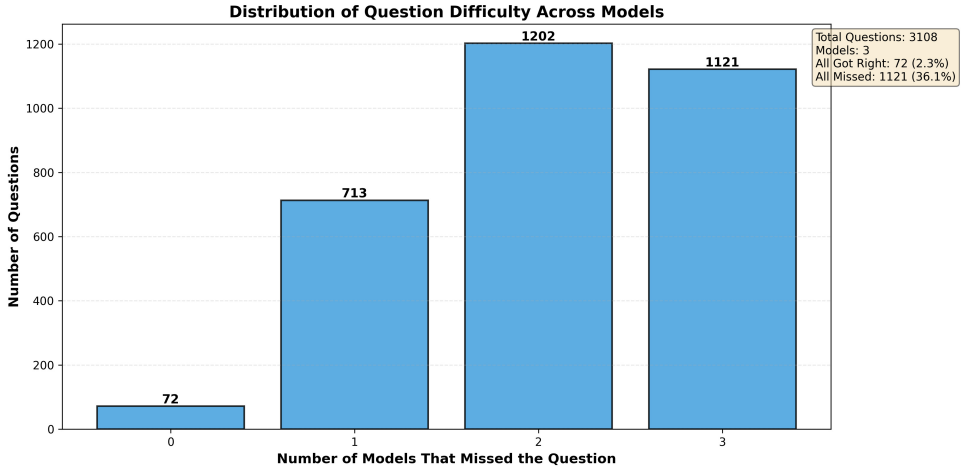


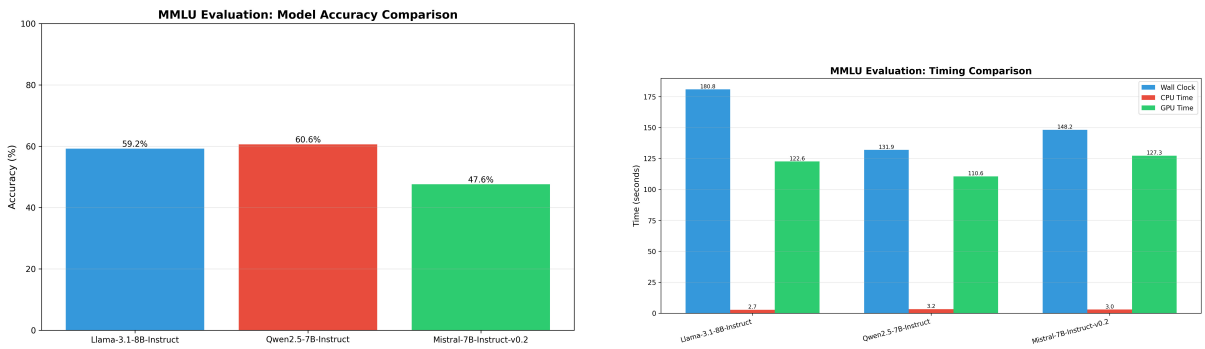
Figure 2: Distribution of missed questions. The high bar on the right (3 models missed) indicates high overlap in failure cases.

3 Part 2: Colab Evaluation (Medium Models)

Larger models (7B-8B parameters) were evaluated using Google Colab's A100 GPU infrastructure.

3.1 Results

There was a marked improvement in accuracy compared to the local run, with Qwen2.5-7B and Llama-3.1-8B both achieving approximately 60% accuracy.



(a) Accuracy Comparison

(b) Timing Comparison

Figure 3: Benchmarking results for 7B/8B parameter models on Colab.

Timing Analysis: Qwen2.5-7B proved to be the most efficient, delivering the highest accuracy (60.6%) with the lowest inference time (132s), outperforming Llama-3.1-8B which took 181s for slightly lower accuracy.

4 Part 3: Code Implementation

4.1 Benchmarking with Timing

The evaluation script was modified to capture granular timing metrics. The code below demonstrates the logic used to separate CPU overhead from GPU inference time using 'torch.cuda.Event'.

```python
def get_model_prediction(model, tokenizer, prompt):
    """Get model's prediction for multiple-choice question with timing"""
    cpu_start = time.process_time()
    inputs = tokenizer(prompt, return_tensors="pt").to(model.device)
    cpu_time_seconds = time.process_time() - cpu_start

    gpu_time_seconds = 0
    gpu_start_event = None
    gpu_end_event = None
    if torch.cuda.is_available() and model.device.type == "cuda":
        torch.cuda.synchronize()
        gpu_start_event = torch.cuda.Event(enable_timing=True)
        gpu_end_event = torch.cuda.Event(enable_timing=True)
        gpu_start_event.record()

    with torch.no_grad():
        outputs = model.generate(
            **inputs,
            max_new_tokens=MAX_NEW_TOKENS,
            pad_token_id=tokenizer.eos_token_id,
            do_sample=False,
            temperature=1.0
        )

    if gpu_start_event and gpu_end_event:
        gpu_end_event.record()
        torch.cuda.synchronize()
        gpu_time_seconds = gpu_start_event.elapsed_time(gpu_end_event) / 1000.0

    return answer, cpu_time_seconds, gpu_time_seconds
```

Listing 1: Timing Logic in llama_mmlu_eval.py

4.2 Verbose Output Logging

To enable better analysis of model failures, I added a logging feature. The snippet below shows the logic that captures the specific question, the choices, the model's answer, and whether the result was correct or incorrect, writing it to a file.

```python
# Write verbose output if enabled
if verbose_output and verbose_file_handle:
    verbose_file_handle.write(f"\n{'='*70}\n")
    verbose_file_handle.write(f"Subject: {subject} | Question {total}\n")
    verbose_file_handle.write(f"{'='*70}\n")
    verbose_file_handle.write(f"Q: {question}\n\n")
    for i, choice in enumerate(choices):
        verbose_file_handle.write(f"  {['A','B','C','D'][i]}. {choice}\n")
    verbose_file_handle.write(f"\nModel Answer: {predicted_answer}\n")
    verbose_file_handle.write(f"Correct Answer: {correct_answer}\n")
    verbose_file_handle.write(f"Result: {'CORRECT' if is_correct else 'WRONG'}\n")
    verbose_file_handle.flush()
```

Listing 2: Verbose Logging Logic

4.3 Chat Agent Context Management

To handle infinite conversation lengths, I implemented a **Summarization Strategy**. When the context window exceeds 6000 tokens, the agent condenses the "middle" of the conversation history while preserving the system prompt and the immediate recent context.

```python
SUMMARIZATION_PROMPT = "Summarize the previous provided conversation history in up to 2000 words. Capture high level objective of conversation and key details."
SUMMARY_THRESHOLD = 6000

def manage_context(chat_history):
    current_tokens = count_tokens(chat_history)
    if current_tokens > SUMMARY_THRESHOLD:
        print("Hit summary threshold.")
        # We need: System (1) + at least 1 message to summarize + Recent (5) = 7 messages
        if len(chat_history) >= 7:
            print("Context threshold exceeded. Summarizing older history...")
            # Summarize everything between system prompt and last 5 messages
            summary_content = get_summary(chat_history[1:-5])

            # Construct new history: [System, Summary, Recent Messages]
            new_history = [chat_history[0]]
            new_history.append({
                "role": "user",
                "content": summary_content
            })
            new_history.extend(chat_history[-5:])
            return new_history
    return chat_history
```

Listing 3: Context Summarization Logic

4.4 Stateless Mode

I implemented a `DISABLE_HISTORY` flag for stateless turns.

```python
# If history is disabled, we only use system prompt and current user input
if DISABLE_HISTORY:
    messages_to_send = [chat_history[0], chat_history[-1]]
else:
    messages_to_send = chat_history

input_ids = tokenizer.apply_chat_template(
    messages_to_send,
    add_generation_prompt=True,
    return_tensors="pt"
).to(model.device)
```

Listing 4: Stateless Mode Logic

4.4.1 Stateless vs Persistent Demonstration

In stateless mode, the agent fails to remember the user's name:

- **User**: My name is Blaise.
- **User**: What is my name?
- **Assistant**: I don't retain any information about our previous conversations.

In persistent mode, the agent correctly identifies the user:

- **Assistant**: I remember! Your name is Blaise.